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Course code : CSA0311

Course name : Data Structures.

C02- Assignment Tool - 1 - Analytical Problem
Solving

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1) Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

Sol:

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *prev, *next;
};

struct Node *head = NULL;

void insert (int value) {
    struct Node *NewNode = (struct Node *) malloc (sizeof (struct Node));
    NewNode->data = value;
    NewNode->next = NULL;
    NewNode->prev = NULL;
    if (head == NULL) {
        head = NewNode;
    }
    return;
    struct Node *temp = head;
    while (temp->next != NULL)
        temp = temp->next;
}

void deleteNode (int value) {
    struct Node *temp = head;
```

```
while (temp != NULL && temp->data != value)
```

```
temp = temp->next;
```

```
if (temp == NULL) {
```

```
printf ("%d not found\n", value);
```

```
return;
```

```
}
```

```
if (temp->prev == NULL)
```

```
temp->prev->next = temp->next;
```

```
else
```

```
head = temp->next;
```

```
if (temp->next != NULL)
```

```
temp->next->prev = temp->prev;
```

```
free(temp);
```

```
printf ("%d deleted\n", value);
```

```
}
```

```
void display()
```

```
{ struct Node *temp = head;
```

```
printf ("NULL<-");
```

```
while (temp != NULL) {
```

```
printf ("%d <->", temp->data);
```

```
temp = temp->next;
```

```
}
```

```
printf ("Null\n");
```

```
}
```

```
int main()
```

```
{ insert (150);
```

```
insert (250);
```

```

insert (350);
insert (450);
insert (550);

printf ("Before Deletion:\n");
display();

delete Node (150);
delete Node (250);

printf ("After deletion:\n");
display();

return 0;

```

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2.) Mr. XYZ booked a ticket on random express. He reached the station and was looking for his PNR number in the chart. Tell out binary search procedure to search the PNR number 4056786. The ticket numbers are 4056779, 4056780, 4056781, 4056782, 4056783, 4056784, 4056785, 4056786, 4056787, 4056788, 4056789.

Ans:

```
#include <stdio.h>
```

```
int binarysearch (int arr[], int n, int target) {
```

```
    int low = 0, high = n-1, steps = 0;
```

```
    while (low <= high) {
```

```
        steps++;
```

```
        int mid = (low+high)/2;
```

```
printf("Step %d: low = %d high = %d mid = %d arr[mid] = %d\n",  
      steps, low, high, mid, arr[mid]);
```

```
if (arr[mid] == target) {
```

```
    printf("Found at index %d in %d steps\n", mid, steps);
```

```
    return mid; }
```

```
else if (arr[mid] < target) {
```

```
    low = mid + 1; }
```

```
else {
```

```
    high = mid - 1; }
```

```
}
```

```
printf("not found\n");
```

```
return -1;
```

```
}
```

```
int main() {
```

```
    int arr[] = {4056779, 4056780, 4056782, 4056783,
```

```
                4056784, 4056785, 4056786, 4056788, 4056789};
```

```
    int n = sizeof(arr) / sizeof(arr[0]);
```

```
    int target = 4056786;
```

```
    binarySearch(arr, n, target);
```

```
    return 0;
```

```
}
```